Homework 2: Novel View Synthesis

Foundations of Spatial AI, Sapienza University of Rome
2/2 Homeworks (total 3 points) Deadline: 3rd November 2026 - 23:59 CET

Throughout the lectures on Neural Rendering, we saw how a radiance field (encoded as a NeRF or a set of 3D Gaussians) is optimized so that its renderings match a set of posed photos, and how it can then render the scene from viewpoints that were never photographed. In this homework you will find out, on a scene of your own, how far these novel views can be trusted and what the radiance field has actually learned.

We suggest Brush[1] (`https://github.com/ArthurBrussee/brush`), which reads a COLMAP reconstruction directly. If you have an NVIDIA GPU with CUDA, you can use nerfstudio (`https://docs.nerf.studio`) instead. Use the same number of training steps in every run, so that your experiments are comparable.

Capture

We suggest you choose a scene that ALSO contains interesting materials and/or several light sources. Examples would be like glossy or metallic objects, lamps, led strips and so forth.



An example of an interesting scene. It includes several objects with different materials, mediums, and partially non-Lambertian surfaces.

You can take about 40 images. Novel view synthesis needs a much broader coverage of the scene than structure-from-motion does, so spend your images on viewpoints that see the scene from different directions rather than on many similar shots.[2] Copy the original files to your computer, as in Homework 1.

If your phone allows it, we suggest locking exposure and white balance. You are not required to: if you do not, look for what changes in the result, and explain it.

Run COLMAP **once** on the full set of images to obtain the camera poses. All the experiments below reuse these poses. Before training, set aside a fixed *test set* of about 5 images spread over the capture, which are never used for training. The same test set is used in every experiment.

[1] Brush: 3D Gaussian Splatting that runs on most GPUs, including laptops and Macs.

[2] You can find useful tips on how to record images at `https://developer.playcanvas.com/user-manual/gaussian-splatting/creating/taking-photos/`

PSNR: Metric for NVS

To evaluate NVS, we need a way to measure how similar a rendered view $\hat{I}$ is with respect to the input real photo I. Although several metrics track this difference, we suggest using *Peak Signal-to-Noise Ratio* (PSNR). Assuming pixel values live in $[0, 1]$, PSNR is computed by expressing on a logarithmic scale the *Mean Squared Error* (MSE) over all N pixels and three color channels:

$$\mathrm{MSE} = \frac{1}{3N} \sum_{\mathbf{p}=1}^{N} \sum_{c=1}^{3} \left(\hat{\mathbf{I}}_c(\mathbf{p}) - \mathbf{I}_c(\mathbf{p}) \right)^2, \qquad \mathrm{PSNR} = 10 \log_{10} \frac{1}{\mathrm{MSE}} \ \mathrm{dB}.$$

An higher value indicates higher photometric similarity. Given that the scale is logarithmic, a +10 dB indicates a ten times smaller error and +3 dB means approximately half the error. As a reference, good 3D Gaussian Splatting reconstructions of benchmark scenes reach roughly 25–30 dB on test views, while training views are usually higher. PSNR is computed per image and then averaged over the training or the test set.

What to do

Baseline Train on all the remaining images (around 35). Report the PSNR on the training images and on the test images, and explain the difference. Then move the camera in the viewer and render one view *between* two training cameras and one view *outside* the region covered by your capture, for instance from below or from far behind the scene. Compare the two.

How many views are enough?

Select a subset of the training images (i.e. 3/4, or 1/2) and retrain the radiance field using only this set. Report the test PSNR on the run and show the same test view rendered by each. *We expect at least 1 experiment.* Be careful: removing images must not change which images are in the test set.

Geometric Distribution of Radiance A radiance field is optimized to reproduce the training images, not to recover the geometry of the scene. Move the camera through the reconstruction and look out for geometric artifacts. These typically happen near or on top of non-lambertian, low textured, or badly covered areas. Show at least one viewpoint where the reconstruction is wrong and explain why the optimization produced it. *We expect at least 1 experiment.*

Optional If you are curious, look at where a mirror or a strong reflection ends up in the scene, or train the baseline again with view-dependent color disabled (in Brush, `--sh-degree 0`) and compare the glossy surfaces. If you have CUDA, train the baseline with both `nerfacto` (a more advanced and optimized reimplementation of NeRF) and `splatfacto` (a more advanced and optimized reimplementation of 3D Gaussian Splatting) in nerfstudio and compare how each one fails.

Report

Hand in a PDF report with **at most** two pages of text or around 6,000 characters; tables and figures do not count toward this limit. *Do not forget your name, surname, and student ID.* Describe the scene and how you photographed it, then for each experiment report the training setup, the PSNR values, and renderings that support your explanation. State the tool you used, the number of training steps, and any setting you changed. Submit the `pdf` report, a `zip` file of all your *original* images (not compressed!), and a short fly-through video (10-20 seconds) of your baseline reconstruction `mp4`. You can record it with `ns-render` in nerfstudio, or with a screen recording of the viewer. Do not zip `pdf` and `mp4`, only images!